Single Pulse Response (SPR) Tab Manual

Overview

The **SPR Tab** allows you to analyze the "Single Pulse Response" (washout characteristics) of your laser ablation system. By analyzing the shape of individual laser pulses, the software can calculate the **Wait Time** (washout time) required for the signal to decay to a specific baseline level (10% or 1%).

This specific washout time is a critical input for the **Optimiser**, ensuring that subsequent laser shots do not overlap unintentionally or that necessary delays are added between spots to prevent signal mixing.

User Interface

The interface is divided into two main sections: **Controls & Results** (Left) and **Visualisation** (Right).

1. Left Panel: Controls & Results

Global Controls

- **Reload Data from iolite**: Refreshes the input data from the main iolite interface. Use this if you have changed selections or processing in iolite while the plugin is open.

Peak Detection

Configure how the software identifies individual laser pulses in the signal.

- **Select Channel**: Choose the isotope channel to analyze (e.g., `U238`, `Th232`). Usually, a high-abundance isotope is best for characterizing washout.
- **Peak Cutoff (Prominence)**:
- **Auto**: Automatically estimates a threshold to identify peaks above the baseline noise.

- **Manual**: Uncheck "Auto" to set a specific prominence value. Higher values filter out noise but may miss smaller peaks.
- **Min. Distance**: The minimum number of data points required between two peaks. Increasing this helps prevent detecting multiple points on the same peak as separate events.
- **Auto-Rescale Y**: Defaults to Checked. When enabled, the plot always scales to show the full pulse intensity.
- **Time Unit**: Toggle between **Seconds (s)** and **Milliseconds (ms)** for display.

SPR Analysis Results

Displays aggregate statistics for all detected (and included) peaks.

- **FW 0.1M (10%)**: Width of the peak at 10% of its maximum height. This is the standard definition for washout time in many applications.
- **Average**: Mean duration across all peaks.
- **RSD**: Relative Standard Deviation (%) – measures consistency.
- **Max**: The longest duration observed (conservative estimate).
- **Area**: Mean integrated area (counts) under the peaks.
- **FW 0.01M (1%)**: Width at 1% of maximum height (stricter washout criterion).

Actions:

- **Apply Average to Optimiser**: Sends the *Average* 10% washout time to the Optimiser tab for the selected channel.
- **Apply Max to Optimiser**: Sends the *Maximum* 10% washout time (more conservative) to the Optimiser tab.
- **Apply Composite to Optimiser**: Sends the calculated width of the **Composite Peak** (average pulse shape) to the Optimiser. This is often the most representative and stable value for washout.

2. Right Panel: Visualisation

SPR Plot (Top)

- Displays the raw signal intensity over time.
- **Markers**:
- **Orange X**: Detected peaks.
- **Grey X**: Excluded peaks.
- **Interaction**: Click on a peak marker or its label to **Exclude/Include** it from the statistics. Excluded peaks are grayed out and removed from calculations.

Composite Plot (Bottom Right - if enabled)

- Shows the "Average Peak Shape" by stacking all valid pulses.
- Useful for visualising the tailing behavior and verifying if the 10% or 1% widths are appropriate.

Mouse Interactions (Plots)

- **Zoom**: Use the **Mouse Wheel** over any plot to zoom in/out on the X-axis. If "Pan/Zoom Y" is enabled, it will also zoom the Y-axis.
- **Pan**: **Left-Click & Drag** a plot to move the view.
- **Double-Click**: Instantly resets the plot to show the full data range.
- **Auto-Rescale Y**: When checked, the plot automatically snaps to show the full pulse intensity whenever values change. Uncheck this to manually zoom into specific features (like the peak tail).

Results Table (Bottom)

- Lists every detected peak individually.
- Columns: `Peak ID`, `Time`, `10% Width`, `10% Area`, `1% Width`, `1% Area`, `Max Intensity`.
- **Interaction**: Rows corresponding to excluded peaks are shown with strikethrough text.

Workflow Guide

Step 1: Load Data

Ensure you have a selection in iolite that contains single laser pulses (e.g., a "Test" or "Tune" line with low repetition rate, like 1 Hz).

1. Open the **Advanced Spot Optimiser**.
2. Click **Reload Data from iolite** if your data is not visible.

Step 2: Select Channel

1. Go to the **SPR** tab.
2. In the **Peak Detection** group, select a representative channel (e.g., `U238` or a matrix element).

Step 3: Optimize Detection

1. Observe the **SPR Plot**. Are all pulses marked with an orange `x`?
2. If noise is being detected as peaks:

- Increase the **Peak Cutoff** (uncheck Auto).
    - Increase **Min. Distance**.
3. If peaks are missed:
    - Decrease the **Peak Cutoff**.

Step 4: Refine Data

1. Look at the **Results Table** or **SPR Plot** for outliers (e.g., double pulses, noise spikes).
2. **Exclude Outliers**: Click on the peak marker in the plot to remove it from the calculation.
    - The statistics in the Left Panel will update instantly.
    - The peak will turn grey in the plot and strikethrough in the table.

Step 5: Apply to Optimiser

1. Decide whether you want to use the **Average**, **Maximum**, or **Composite** washout time.
    - *Average*: Good for stable systems.
    - *Max*: Safer to ensure zero overlap.
    - *Composite*: Best for noisy signals or when you want the most statistically representative shape.
2. Click **Apply Average/Max/Composite to Optimiser**.
3. A confirmation message will appear. This value is now set as the "Custom" dwell/wait time for this channel in the **Optimiser Tab**.

Troubleshooting

- **No Peaks Detected**:
- Check if the correct channel is selected.
- Lower the **Peak Cutoff**.
- Ensure your data actually contains single pulses.
- **"Error" in Results**:
- Usually means the baseline is too high or the pulse never drops below 10%/1% within the available window.
- Try selecting a cleaner channel.
- **Optimiser vs SPR Units**:
- The Optimiser usually works in **Milliseconds**. the SPR tab handles unit conversion automatically when you click "Apply".

Methodology & References

The **SPR Tab** characterizes system washout by analyzing the temporal profile of individual laser pulses.

Peak Analysis Process:

1. **Detection**: Peaks are identified using a prominence-based algorithm.
2. **Width Calculation**: The software determines the time taken for the signal to decay from its maximum to 10% (FW 0.1M) and 1% (FW 0.01M) of its peak height.
3. **Composite Peak**: An "Average Peak Shape" is generated by time-aligning and stacking all valid pulses. This provides a high-signal-to-noise profile for precise washout determination, even on trace elements.